Soroush Piri

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TECHNICAL SKILLS

- **Programming & Data Science:** Python, R, SQL, SQLite, Pandas, NumPy, Scikit-learn, XGBoost, Tidyverse
- **AI & LLM Systems:** Agentic AI Systems, Large Language Models (LLMs), Conversational AI, AI Orchestration, Prompt Engineering, Tool-Calling Architectures
- **Machine Learning & Analytics:** Predictive Modeling, Time-Series Forecasting, Clustering (K-Means), Feature Engineering, Regression Analysis, Ensemble Learning
- **Cloud & Data Infrastructure:** AWS Bedrock, Amazon S3, Boto3, ETL Pipelines, SQL Query Optimization, Database Indexing
- **Visualization & Interactive Systems:** Streamlit, R Shiny, Plotly, Folium, Geospatial Analysis, Interactive Dashboards
- **NLP & Text Analytics:** Natural Language Processing (NLP), Text Mining, PDF Parsing, Information Extraction, LLM-Assisted Analytics

WORK EXPERIENCE

AI / Data Science Intern

May 2026 – Aug 2026

ProAssurance Group Services Corporation

- Developed an AI-powered conversational analytics agent using AWS Bedrock, Python, and SQLite to analyze large-scale NPI healthcare provider datasets and support insurance risk assessment workflows.
- Engineered an LLM-integrated AI agent capable of performing intelligent provider search, taxonomy analysis, geospatial mapping, and contextual follow-up reasoning across healthcare datasets.
- Built a healthcare data AI assistant with tool-calling architecture, SQL-based retrieval pipelines, and conversational memory to generate actionable insights for analytical and business decision-making.
- Designed and deployed an interactive AI analytics platform using Streamlit, AWS Bedrock, SQLite, and geospatial visualization tools for provider intelligence and healthcare network analysis.
- Developed scalable ETL and indexing pipelines for millions of healthcare provider records to optimize AI-agent query performance and large-scale data processing.
- Implemented contextual memory and conversational history capabilities within the AI agent to support multi-step analytical workflows and dynamic follow-up questioning.

Graduate Researcher - AI and Machine Learning

Aug 2023 – May 2026

School Of Architecture and Planning, Morgan State University, Baltimore, MD

- Architected a hybrid NLP pipeline combining transformer-based semantic modeling with a multi-state extraction engine to convert unstructured construction fatality reports into structured, research-ready datasets.
- Designed a scalable research framework for extracting, standardizing, and analyzing occupational fatality data, establishing a methodological foundation for AI-driven safety modeling.
- Leveraged embeddings and semantic similarity to consolidate fragmented contributing-factor labels into a standardized taxonomy, enabling large-scale causal analysis in construction accidents.
- Engineered an interactive R Shiny platform integrating OCR, rule-based parsing, and LLM-assisted extraction with validation tooling to support scalable accident-report analysis.
- Applied clustering, PCA, and time-series ML to large-scale infrastructure data, developing robust evaluation metrics that revealed deterioration patterns and informed DOT maintenance strategy.
- Led an end-to-end ML evaluation initiative on safety and crime data, designing scalable frameworks for preprocessing, model performance assessment, and policy insight generation for Maryland state initiatives.
- Built multimodal evaluation dashboards (R Shiny, Power BI, SQL, Python/R integration) to translate complex ML outputs into real-time decision support for policymakers.

Graduate Research Assistant

Aug 2021 - May 2023

Urban Integrity and Intelligence Lab, Southern Illinois University, Carbondale, IL

- Consulted with faculty and external partners on predictive modeling and evaluation design, delivering statistical frameworks and performance metrics that guided multiple ML research projects.
- Built SQL queries and visualization pipelines to generate evaluation dashboards capturing trends, business performance, and model reliability.
- Developed statistical and ML evaluation methods for data collection, quality assurance, and analysis, ensuring reproducible and scalable results.
- Designed and deployed interactive R Shiny applications that implementedend-to-end evaluation workflows from raw data ingestion to cleaning, model assessment, and visualization for end users.
- Educated participants on end-to-end automation frameworks and deployment strategies for open-source data.
- Demonstrated reproducible workflows for multimodal data evaluation (text, tabular, visualizations), enabling researchers to apply frameworks to real-world datasets.

APPLICATIONS AND TOOLS DEVELOPED

IntegrativeSentiment.io (iS.io)

Aug 2021 - May 2023

A Web-based Interactive Tool for Integrated Human Sentiment Modeling

- Featured and demonstrated IS.io app at SIMAUD 2022 as a novel evaluation framework for analyzing large-scale human feedback through sustainability and sentiment modeling studies.
- Designed and deployed a real-time NLP evaluation platform in R Shiny for analyzing large-scale textual data (Twitter), integrating transformer-based NLP and custom performance metrics.
- Developed an interactive browser-based interface embedding the R engine, enabling scalable evaluation without reliance on RStudio or command-line tools.
- Integrated graph theory and geospatial mapping to create a multimodal evaluation system that assessed sentiment patterns across time and space.
- Applied advanced emotional intelligence and NLP frameworks to generate nuanced evaluation metrics beyond positive/negative sentiment, capturing complexity in human feedback.

EDUCATION

PhD Candidate - Concentration: Data Engineering, Machine Learning and Artificial Intelligence - GPA:3.8

Aug 2023 - May 2027

Morgan State University, Baltimore, MD

Master of Architecture (M.Arch) - GPA:3.61

Aug 2021 - May 2023

Southern Illinois University, Carbondale, IL

Bachelor of Science (BS) In Architecture - GPA:3.52

Sep 2015 - Jan 2020

Tehran Azad University, Tehran, Iran

PUBLICATION

Google Scholor: https://scholar.google.com/citations?user=2G7H_DMAAAAJ&hl=en

HONORS AND AWARDS

Machine Learning Framework for Predicting Deterioration of Concrete Bridges in Maryland — April 2026

Structures Congress 2026 Journal- Accepted and Published Paper

Uncovering Symmetric and Asymmetric Deterioration Patterns in Maryland's Steel Bridges Through Time-Series Clustering and Principal Component Analysis — November 2025

Symmetry Journal - Accepted and Published Paper

Development of Bridge Deterioration Models Using Machine Learning Methods — August 2025

CIAMTIS – U.S. DOT Region 3 University Transportation Center – Published Project

Integrating MachineLearning Techniques for Enhanced Safety and Crime Analysis in Maryland — April 2025

Applied Sciences Journal -Novel Applications of Machine Learning and Bayesian Optimization - Accepted and Published Paper

Unraveling Safety Concerns in Construction: A ComprehensiveData Analysis — May 2024

ASC 2024 International Conference in Alabama - Accepted and Published Paper

ExploringU.S.Occupant PerceptionTowardIndoor Air Quality Via Social Media and NLPAnalysis — May 2024

Journal of Environmental Scienceand Public Health 2024 - Accepted and Published Paper

PATENT

Patent Pending - Agentic AI Framework for Scalable Data-Driven Decision Support Using LLM-Driven Analytical Pipelines

Soroush Piri, Dr. Zeinab Bandpey, Dr. Mehdi Shokouhian

- Developed an agentic AI framework integrating AWS Bedrock large language models with constrained analytical tool execution for reliable, interpretable, and scalable decision-support workflows across large-scale infrastructure datasets.
- Designed structured AI orchestration pipelines that transform natural language queries into validated analytical execution workflows using intent extraction, tool routing, schema validation, and deterministic backend computation.
- Built context-aware conversational AI systems with session-state memory, follow-up reasoning, and subset-aware analytical execution for multi-step infrastructure analysis and decision support.
- Integrated machine learning forecasting pipelines, time-series clustering, deterioration modeling, and statistical analytics within an LLM-driven analytical execution architecture.
- Engineered deterministic backend analytical modules for forecasting, K-Means clustering, trend analysis, feature engineering, uncertainty estimation, and bridge deterioration assessment using FHWA NBI and LTBP datasets.
- Architected a grounded AI reasoning framework separating LLM orchestration from numerical computation to improve transparency, interpretability, and reduction of hallucinated analytical outputs.